

Hot Work Permit Issuance Procedure

1. Purpose

This Standard Operating Procedure (SOP) defines the controlled process for issuing, using, suspending, and closing hot work permits in a petrochemical plant. The procedure is intended to prevent fires, explosions, personnel exposure, and loss of containment by ensuring that hot work is performed only after hazards are identified, controls are established, atmospheric conditions are verified, and responsible personnel understand their duties.

2. Scope

This SOP applies to welding, cutting, brazing, grinding, soldering, heating, and other activities capable of producing flame, sparks, hot surfaces, or sufficient heat to ignite combustible materials. It applies to employees, contractors, maintenance teams, operations personnel, permit issuers, fire watches, and area supervisors working within controlled plant areas. The procedure shall be used together with site isolation, lockout/tagout, confined-space, gas-testing, and emergency-response requirements.

3. Roles and Responsibilities

3.1 Permit Issuer: The permit issuer shall verify the work location, job scope, hazards, required isolations, fire-prevention controls, and atmospheric test results. The issuer shall ensure that the permit is complete, communicated, displayed, and authorized before work starts. The issuer shall stop or suspend the permit when conditions become unsafe.

3.2 Fire Watch: The fire watch shall remain assigned to the work area, continuously observe for ignition, sparks, smoke, or smoldering materials, maintain suitable firefighting equipment, and raise the alarm when required. The fire watch shall not perform unrelated duties that interfere with continuous monitoring.

3.3 Area Supervisor: The area supervisor shall coordinate with operations and maintenance, confirm area readiness, ensure workers understand the permit controls, verify required equipment and barriers, and maintain control of the work area. The supervisor shall ensure the job stops when permit conditions are no longer satisfied.

4. Pre-Work Checklist

4.1 Confirm the exact job location, equipment identification, work description, and responsible work crew.

4.2 Review process hazards, nearby hydrocarbons, combustible materials, drains, vents, open lines, tanks, and adjacent operations.

4.3 Verify required equipment isolation, depressurization, draining, purging, blinding, lockout/tagout, and line clearance according to the approved work plan.

4.4 Remove combustible materials from the hot work zone or protect them with suitable fire-resistant covers. Protect cables, hoses, insulation, scaffolding, and nearby equipment from sparks and heat.

4.5 Confirm suitable fire extinguishers, fire hoses, barriers, screens, communication equipment, and emergency access.

4.6 Check welding machines, leads, holders, torches, regulators, cylinders, flashback protection, grounding, and other equipment for safe condition.

4.7 Establish a defined hot work boundary and prevent unauthorized entry.

4.8 Confirm that the fire watch is briefed, equipped, and available for the required duration.

4.9 Review simultaneous operations and confirm that nearby work will not create an incompatible hazard.

5. Gas Testing Requirements

5.1 Atmospheric testing shall be performed by an authorized and competent gas tester using calibrated and suitable instruments.

5.2 Test the atmosphere at representative locations, including the work zone, low points, nearby drains, pits, enclosed spaces, and other areas where flammable vapors may collect.

5.3 Test for oxygen concentration and flammable vapors, and test for toxic gases when identified by the hazard assessment or process conditions.

5.4 The atmosphere shall meet the plant's approved hot work limits before permit issuance. As a common control, flammable vapor concentration must remain below the site's specified percentage of the lower explosive limit, with the plant standard taking precedence.

5.5 Record the test time, location, instrument identification, readings, and tester name on the permit.

5.6 Repeat testing at intervals defined by the site's risk assessment, and whenever conditions change or work is interrupted. Continuous monitoring shall be used where required by the hazard assessment.

6. Permit Issuance and Validity

6.1 The permit issuer shall review all checklist items and authorize the permit only after required controls are verified.

6.2 The permit shall identify the work location, job, hazards, controls, gas-test results, responsible personnel, issue time, and expiry time.

6.3 The permit is valid only for the specified job, location, conditions, and period. Unless the plant's permit system

specifies otherwise, a hot work permit should be limited to one work shift and shall not exceed 12 hours without formal revalidation and reauthorization.

6.4 A new or revalidated permit shall be required after expiry, significant work-scope change, relocation, loss of required controls, or a change in plant conditions.

6.5 The permit shall be available at the work location and closed when the job is completed or stopped.

7. Conditions for Permit Suspension

The permit shall be suspended immediately when gas-test results exceed approved limits, hydrocarbon or chemical release is suspected, fire protection becomes unavailable, required isolation is compromised, weather or wind creates an unsafe condition, combustible material enters the hot work zone, alarms or emergency conditions occur, the fire watch leaves, or the work scope changes. Work shall not resume until the area is reassessed, controls are restored, required gas testing is repeated, and the permit is formally reauthorized.

8. Emergency Shutdown Triggers

Hot work shall stop immediately and the area shall enter the applicable emergency response process when there is a fire, explosion, uncontrolled release, rapidly increasing flammable-gas reading, loss of containment, confirmed hazardous-gas alarm, loss of critical isolation, major equipment failure, or any condition presenting immediate danger to personnel or plant integrity. Personnel shall raise the alarm, withdraw to the designated safe location, and follow site emergency procedures. Do not attempt firefighting beyond the training, equipment, and authorization provided.

9. Completion and Closeout

After work, the area supervisor shall confirm that hot surfaces, sparks, slag, and smoldering materials are eliminated. The fire watch shall continue monitoring for the period specified by the site risk assessment, especially where insulation, concealed spaces, or combustible materials could retain heat. The permit issuer shall verify housekeeping, remove temporary controls only when authorized, record completion, and formally close the permit.

10. Documentation and Records

Retain the completed permit, gas-test records, isolation references, revalidation entries, suspension details, and closeout information according to plant document-control requirements. Any abnormal condition, incident, near miss, or repeated permit suspension shall be reported through the plant's incident-management process and reviewed by the responsible supervisor and safety personnel.